Supplementary Material

Proofs for "Separating Numerical Noise from Thermodynamic Inconsistency in Cycle-Consistency Tests on Metabolic Circuits"

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This document collects the proofs of every numbered result in the main text, in order of appearance. Section Sk here proves the result whose proof the main text defers to Section Sk. Notation, numbering, and equation references follow the main text.

S1

(i) The net rate of change of c_i is incoming minus outgoing conversion rate; under (3) the conversion rate along (i, j) is J_{ij}, so stationarity of c_i is exactly the vanishing of the signed sum, which is the current law at node i.

(ii) The sum telescopes to $\mu_{i_1} - \mu_{i_1} = 0$ identically, because μ is a function on vertices. The substantive content is the converse: a set of rate constants admits a consistent potential function μ if and only if the products of forward rate constants around every cycle equal the products of reverse constants — the Wegscheider condition (Schuster & Schuster, 1989; Wegscheider, 1901). A network violating it admits no μ and hence no circuit representation.

S2

$z_\ell \in \ker B$ by Definition 2.6, so $Bz_\ell = 0$ and hence $H_\ell(\mu) = z_\ell^\top B^\top \mu = (Bz_\ell)^\top \mu = 0$.

S3

$\operatorname{im} B^\top = (\ker B)^\perp = \mathcal{Z}(G)^\perp$ by the fundamental theorem of linear algebra. Hence $d \in \operatorname{im} B^\top$ iff $z^\top d = 0$ for all $z \in \mathcal{Z}(G)$, and since a basis spans $\mathcal{Z}(G)$, iff $z_\ell^\top d = 0$ for every ℓ in that basis.

S4

Each difference $\Delta\mu_e = \mu_j - \mu_i$ is computed with one rounding: $\mathrm{fl}(\mu_j - \mu_i) = (\mu_j - \mu_i)(1 + \theta_e)$, $|\theta_e| \le u$, giving an absolute error at most $u|\mu_j - \mu_i| \le 2u\Lambda_\ell$. Summing L such quantities sequentially, the standard forward error bound for floating-point summation (Higham, 2002, Ch. 4) gives a total error at most $\gamma_L \sum_{e\in\ell} |\Delta\mu_e|$ from the additions, plus the propagated per-difference errors, at most $\sum_e u|\Delta\mu_e|$. Since $|\Delta\mu_e| \le 2\Lambda_\ell$ and there are L edges, $\sum_{e\in\ell} |\Delta\mu_e| \le 2L\Lambda_\ell$. Collecting the two contributions and absorbing the constants into γ_{2L} yields (7). Because the exact value is 0, the computed value equals its own error, which is what the display asserts.

For the compensated variant, Kahan summation (Higham, 2002; Kahan, 1965) replaces the γ_L factor by $2u + O(Lu^2)$, leaving only the L per-difference roundings, hence $O(Lu\Lambda_\ell)$.

The bound is stated in $\Lambda_\ell = \max|\mu_i|$ only through $|\Delta\mu_e| \le 2\Lambda_\ell$; every step above holds with $2\Lambda_\ell$ replaced by the spread $\max\mu - \min\mu$ on the cycle, which is gauge-invariant. When all potentials on the cycle lie within a factor of two of one another, Sterbenz's lemma (Goldberg, 1991) makes each difference exact, and since each partial sum $\mu_{i_{k+1}} - \mu_{i_1}$ is then itself a difference of two such potentials, every addition is exact as well: the computed cycle sum is exactly zero.

S5

Immediate from (7), whose right-hand side is, for fixed u, proportional to $\gamma_{2L}\, L\Lambda_\ell$ and hence to $L^2\Lambda_\ell$ to first order in Lu. Two cycles whose values of $L^2\Lambda$ differ by a factor r have floors differing by r, and a constant threshold separating noise from signal on both must lie within a factor r of each floor at once, which fails once r exceeds the dynamic range the test requires.

S6

$H_\ell(\mu + c\mathbf{1}) = z_\ell^\top B^\top(\mu + c\mathbf{1}) = H_\ell(\mu) + c\, z_\ell^\top B^\top \mathbf{1} = H_\ell(\mu)$, since $B^\top\mathbf{1} = 0$ (every edge has one head and one tail). The stated c^* centres the interval $[\min\mu, \max\mu]$ on zero, minimising $\max|\mu_i|$, which is the definition of Λ_ℓ.

S7

Lemma 2.17. With $d = B^\top\hat{\mu}$, $H_\ell(d) = z_\ell^\top B^\top\hat{\mu} = (Bz_\ell)^\top\hat{\mu} = 0$, since $Bz_\ell = 0$ by Definition 2.6. The argument holds for every $\hat{\mu}$, and in particular for $\hat{\mu} = \mu + \eta$ with any η. Equivalently, the coefficient of $\hat{\mu}_i$ in the cycle sum is the number of times the cycle enters species i minus the number of times it leaves, which is zero for every species of a closed walk.

Theorem 2.19. With $d = \bar{d} + \eta$ and $\bar{d}$ consistent, $H_\ell(d) = z_\ell^\top\bar{d} + z_\ell^\top\eta = z_\ell^\top\eta$ by Proposition 2.10. This is linear in the independent errors, so it has mean zero and variance $\sum_e z_\ell(e)^2\sigma_e^2$, which is (8); for a simple cycle $z_\ell(e) \in \{-1, 0, 1\}$ and the sum runs over the edges of ℓ. Under normality $z_\ell^\top\eta$ is normal, and $\Pr[|z_\ell^\top\eta| \le z_{1-\alpha/2}s_\ell] = 1 - \alpha$ exactly. Without normality, Chebyshev's inequality gives the same statement with $z_{1-\alpha/2}$ replaced by $\alpha^{-1/2}$.

The two results are the two halves of one fact: the cycle sum reads only the part of the edge data that is not a gradient, so an error enters it if and only if it enters edge by edge.

S8

With exact data, Lemma 2.8 gives $H_\ell = 0$ and Theorem 2.12 bounds the computed value by $\varepsilon_{\text{num}}(\ell) \le \varepsilon^*(\ell)$; the strict inequality in the event is therefore never satisfied. With uncertain edge data, $|H_\ell| \le \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$ by Theorem 2.19, and on that event the computed value satisfies $|\widehat{H}_\ell| \le |H_\ell| + \varepsilon_{\text{num}}(\ell) \le \varepsilon_{\text{data}}(\ell) + \varepsilon_{\text{num}}(\ell) = \varepsilon^*(\ell)$. Hence $\Pr[|\widehat{H}_\ell| > \varepsilon^*(\ell)] \le \alpha$. (With ε^* the maximum of the two floors rather than their sum, the last inequality fails, which is why the sum is used.) Strictly, the rounding bound is here applied to a sum whose exact value is H_ℓ rather than zero; the argument of S4 applies unchanged with $\sum_{e\in\ell}|d_e|$ in place of $2L\Lambda_\ell$, which for edge data within a few σ_e of a gradient changes the bound negligibly.

S9

With exact data, the computed value differs from the true one by at most $\varepsilon_{\text{num}}(\ell) \le \varepsilon^*(\ell)$ (Theorem 2.12), so $|\widehat{H}_\ell| \ge D - \varepsilon^*(\ell) > 2\varepsilon^*(\ell) - \varepsilon^*(\ell) = \varepsilon^*(\ell)$. With independent errors on the edge data, the true sum is $\pm D + z_\ell^\top\eta$, and $|z_\ell^\top\eta| \le \varepsilon_{\text{data}}(\ell)$ with probability at least $1 - \alpha$ (S7). On that event $|\widehat{H}_\ell| \ge D - \varepsilon_{\text{data}}(\ell) - \varepsilon_{\text{num}}(\ell) = D - \varepsilon^*(\ell) > \varepsilon^*(\ell)$.

S10

(i) The three conditions partition $[0, \infty)$ at the two cut points $\varepsilon_{\text{num}} \le \varepsilon^* = \varepsilon_{\text{num}} + \varepsilon_{\text{data}}$. When $\varepsilon_{\text{data}}$ is undefined (Remark 2.20) there is one cut point and every value above it is UNDECIDABLE.

(ii) The middle interval $(\varepsilon_{\text{num}}, \varepsilon_{\text{num}} + \varepsilon_{\text{data}}]$ is non-empty exactly when $\varepsilon_{\text{data}} > 0$. Numerically, the measured noise in a long uncentred high-potential cycle is of order 10^{-12} kJ/mol and the bound ε_{num} at most of order 10^{-9} (main text, Table 1 and Figure 1), while edge-level uncertainties of 1–10 kJ/mol (Flamholz, Noor, Bar-Even, & Milo, 2012; Noor, Haraldsdóttir, Milo, & Fleming, 2013) give $\varepsilon_{\text{data}}$ of order 1–10 kJ/mol. The middle interval therefore spans some nine to thirteen orders of magnitude.

(iii) A two-valued rule places a single cut at some t. If $t < \varepsilon^*$, cycles with $|\widehat{H}| \in (t, \varepsilon^*]$ are called inconsistent though their sums are within what the data's own uncertainty and the arithmetic can produce from a consistent network. If $t \geq \varepsilon^*$, then $t > \varepsilon_{\text{num}}$, and cycles with $|\widehat{H}| \in (\varepsilon_{\text{num}}, t]$ are called consistent though their sums exceed what the arithmetic alone can produce from exact data, so the verdict asserts a consistency the data do not establish. Either way one of the two misstatements occurs.

S11

Each $\mathcal{B}_i$ spans $\mathcal{Z}(G)$ by Definition 2.6. The map $z \mapsto z^\top d$ is linear, so it vanishes on a spanning set if and only if it vanishes on the whole space, and hence if and only if it vanishes on any other spanning set.

S12

Let G be the "theta-plus" graph: vertices $\{a, b\}$ joined by four internally disjoint paths $P_1, \ldots, P_4$, giving $\nu = 3$. Let $e^* \in P_1$ carry a defect of size D, all other edges consistent, so that any cycle containing e^* exactly once has cycle sum $\pm D$ and any cycle avoiding it has sum 0.

Take $\mathcal{B}_1 = \{P_1P_2^{-1},\ P_2P_3^{-1},\ P_3P_4^{-1}\}$. Only the first contains e^*, so exactly one element has nonzero sum.

Take $\mathcal{B}_2 = \{P_1P_2^{-1},\ P_1P_3^{-1},\ P_1P_4^{-1}\}$. This is a basis: the three are independent, since $P_1P_j^{-1}$ is the only element containing P_j, for $j = 2, 3, 4$. Every element contains e^*, so all three have nonzero sum. Both are cycle bases of the same graph and the same data, and they flag one and three cycles respectively.

S13

With the remaining edge data exact, a cycle's true sum is D times its net signed traversal count of e^*, and 0 from every other edge. A cycle with nonzero count has $|H_\ell| \geq D > 2\varepsilon^*(\ell)$ and is flagged by Theorem 2.24; a cycle with zero count has $H_\ell = 0$ and is not flagged, by Theorem 2.23. For simple cycles a nonzero count is containment. Hence every flagged cycle contains e^*, so e^* lies in their intersection, which is $W(\mathcal{B})$. With independent errors on the edge data, each of the ν basis cycles receives the verdict the exact case assigns it except with probability at most α (Theorems 2.23 and 2.24); by the union bound all ν do so with probability at least $1 - \nu\alpha$, and on that event the argument above applies.

S14

By Rule S2 the only introduction form for `Verdict` requires a premise of type `Tolerance`, and the conclusion carries it by Definition S1. A numeric literal has no type in the grammar of Definition S1, so there is no derivation in which one appears in the second premise; and no coercion rule from a numeric literal to `Tolerance` exists, since `Tolerance` is introduced only by the computation of (10) from a `Loop` and its data. Hence no derivation produces a bare verdict.

S15: a language in which the tolerance cannot be omitted

The analysis of the main text is only as good as its implementation. A practitioner who computes ε^* correctly and then writes `if abs(H) > 1e-6` has gained nothing. We therefore express the protocol in a small typed language in which the omission is not expressible.

Types

Definition S1 (Types)**.** The types are

$$\tau \;::=\; \texttt{Circuit} \mid \texttt{Loop} \mid \texttt{Holonomy} \mid \texttt{Tolerance} \mid \texttt{Verdict} \mid \texttt{Witness}.$$

A value of type `Holonomy` is a pair $(\widehat{H}_\ell, \ell)$; a value of type `Tolerance` is a triple $(\varepsilon_{\text{num}}, \varepsilon_{\text{data}}, \varepsilon^*)$ with $\varepsilon_{\text{data}}$ possibly the distinguished value `undefined`; a value of type `Verdict` is an element of $\{\mathsf{CONSISTENT}, \mathsf{UNDECIDABLE}, \mathsf{INCONSISTENT}\}$ *paired with* the `Tolerance` that produced it.

The typing rule that does the work

Rule S2 (Verdict requires a tolerance)**.**

$$\frac{\Gamma \vdash h : \texttt{Holonomy} \qquad \Gamma \vdash t : \texttt{Tolerance}}{\Gamma \vdash \texttt{admit}\ h\ \texttt{tolerance}\ t : \texttt{Verdict}}$$

There is no other rule with `Verdict` in its conclusion.

Theorem S3 (No bare verdict)**.** *No well-typed program produces a value of type* `Verdict` *that is not paired with the* `Tolerance` *used to derive it. In particular, a comparison of a* `Holonomy` *against a numeric literal is not well-typed.*

Proof. See Section S14 above.

Remark S4 (What this does and does not buy). Theorem S3 rules out a class of implementation error: it is not possible to forget the tolerance, to hard-code it, or to report a verdict whose provenance is not recoverable. It does not make the tolerance *correct*: a program supplying a wrong σ_i to (9) is well-typed and wrong. The type system enforces accountability of the computation, not accuracy of the inputs, and we state this because the distinction is easy to overstate in the other direction.

The protocol as a program

Listing 1: The cycle-consistency test with cycle-local tolerances. Every verdict carries its tolerance by Theorem S3; the validity gate of Criterion 2.37 precedes any diagnostic claim.

```
-- kcl-tolerance.cfc
-- Cycle-consistency with per-cycle tolerances.

import Recon3D
import Recon3D.uncertainty as sigma

circuit Reference from Recon3D.glycolysis_tca {
  solve yield C_ref
}

-- Gauge-centre potentials to tighten the bound on the numerical
-- floor (gauge proposition). Changes no cycle sum.
let C_ref_centred := centre_potentials(C_ref)
```

```
-- Minimum-weight cycle basis: sharper witness sets.
let B := minimum_cycle_basis(C_ref_centred)

-- VALIDITY GATE (invalid versus negative). The reference network
-- must pass before any diagnostic verdict is meaningful. Data floors
-- here are taken at level alpha / size(B).
foreach loop in B {
  holonomy of loop in C_ref_centred yield h_ref
  tolerance of loop with sigma yield t
  admit h_ref tolerance t yield v_ref

  assert v_ref != INCONSISTENT
    otherwise invalid
      emit "reference network fails its own consistency check"
}

emit "validity gate passed: reference is self-consistent"

-- DIAGNOSTIC. Perturbed network: the forward constant alone is
-- scaled, which breaks the Haldane relation and injects a defect of
-- RT ln(1/0.15) = 4.9 kJ/mol on the PK edge.
circuit Perturbed from Recon3D.glycolysis_tca {
  reaction PK : PEP -> Pyruvate, k : 0.15 * Recon3D.k_PK
  solve yield C_pert
}

let C_pert_centred := centre_potentials(C_pert)

let flagged := {}
foreach loop in B {
  holonomy of loop in C_pert_centred yield h
  tolerance of loop with sigma yield t
  admit h tolerance t yield v

  report loop, h, t.numerical, t.data, t.star, v

  where v == INCONSISTENT collect loop into flagged
}

-- Localisation is to the witness set, not to a named loop.
witness of flagged yield W
report "witness_set", W
report "witness_cardinality", size(W)

-- The claim under test, stated basis-independently.
assert Recon3D.edge("PK") in W
  emit "defect localised: PK edge in witness set"
  otherwise decline
    emit "defect not localised at this data quality"
```

S16: protocol and pre-registered criteria

Data

1. **Network.** The glycolysis, pentose phosphate, and TCA subnetwork of a human metabolic reconstruction (Brunk et al., 2018), approximately 80 reactions over 60 metabolites.

2. **Edge data.** For each reaction with forward and reverse rate constants in the kinetic models (Mulquiney & Kuchel, 1999; Rapoport & Heinrich, 1975), the edge datum is the standard reaction energy those constants imply by the Haldane relation, $d_e = -RT\ln(k_e^+/k_e^-)$, with its reported uncertainty σ_e; where an equilibrium constant is reported independently for the reaction, it is used instead. Edges whose datum is derived from component-contribution formation energies (Flamholz et al., 2012; Noor et al., 2013) are consistent by construction (Lemma 2.17) and are recorded as such. Edges without a reported uncertainty are recorded and trigger Remark 2.20.

3. **Reference physiology.** Erythrocyte metabolite concentrations and glycolytic flux from established kinetic models (Mulquiney & Kuchel, 1999; Rapoport & Heinrich, 1975).

4. **Perturbation.** The pyruvate kinase forward rate constant reduced to 15% of reference with the reverse constant unchanged, which injects a defect of $RT\ln(1/0.15) \approx 4.9$ kJ/mol on the PK edge. The 15% matches the residual activity documented in the clinical deficiency (Zanella, Fermo, Bianchi, & Valentini, 2005); the deficiency itself scales both directions, which is Control 1.

Procedure

Protocol S5 (Cycle-consistency with cycle-local tolerance)**.**

1. Construct the reference circuit by Definitions 2.2 and 2.3; verify the current law at every internal node to the numerical floor of Theorem 2.12 applied to the node balance.

2. Gauge-centre the potentials (Proposition 2.15) and record the resulting Λ_ℓ per cycle, both before and after.

3. Compute a minimum-weight cycle basis (Horton, 1987); record ν, the length distribution, and the basis itself so that the computation is reproducible. Verify that the PK edge lies on at least one cycle; if it is a bridge, no cycle sum can detect the perturbation and the experiment is invalid by design.

4. For each cycle compute $\varepsilon_{\text{num}}(\ell)$ by (7) and $\varepsilon_{\text{data}}(\ell)$ by (9), at α/ν for the validity gate and at $\alpha = 0.01$ otherwise; record both, and ε^*.

5. **Validity gate.** Evaluate Definition 2.36 on the reference. If it fails, halt and report an *invalid* experiment per Criterion 2.37.

6. Evaluate the perturbed circuit; record $\widehat{H}_\ell$ and verdict(ℓ) for every cycle.

7. Compute the witness set W (Definition 2.32) and record $|W|$ and whether the perturbed edge lies in it.

8. **Control 1 (activity loss).** Repeat step 6 with both PK constants scaled by 0.15, which preserves the Haldane relation and every cycle sum. Any INCONSISTENT verdict here is a false positive; at level α/ν per cycle its probability is at most α (Theorem 2.23). This control also shows directly that the test is blind to a loss of activity that leaves the equilibrium constant unchanged.

9. **Control 2 (basis).** Repeat steps 3–7 with a *different* cycle basis (e.g. a fundamental basis from a spanning tree). Verify the network-level verdict agrees (Theorem 2.29) and record how F and W differ (Proposition 2.30).

10. **Control 3 (detection sweep).** Sweep the perturbation magnitude and record the smallest D at which the correct edge enters W; compare against the predicted $2\varepsilon^*$ of Theorem 2.24.

Pre-registered criteria

Criterion S6 (Success)**.**

(a) **Validity.** The reference network passes Definition 2.36 on every basis cycle.

(b) **No false positives.** Control 1 produces zero INCONSISTENT verdicts, as Theorem 2.23 requires.

(c) **Localisation.** The perturbed edge lies in W, and $|W| \leq 3$.

(d) **Basis invariance.** Control 2 yields the same network-level verdict under both bases.

(e) **Detection threshold.** Control 3 finds the empirical detection threshold within a factor of 10 of the predicted $2\varepsilon^*$.

Criterion S7 (Failure, and what each failure means)**.**

- Failure of (a) renders the experiment *invalid* (Criterion 2.37): the reconstruction's thermodynamic annotations are not self-consistent to their claimed precision. No conclusion about the diagnostic follows.
- Failure of (b) has probability at most α under the stated error model. It indicates an error in the tolerance computation, or reported uncertainties that understate the real errors of the edge data; it is not a detection.
- Failure of (c) with (a) and (b) passing is a genuine *negative*: the test does not localise this defect at this data quality. If additionally most verdicts are UNDECIDABLE, the limiting factor is $\varepsilon_{\text{data}}$ and the finding is about the database (Remark 2.28).
- Failure of (d) would contradict Theorem 2.29 and indicates an implementation error in the basis computation.
- Failure of (e) indicates the bound of Theorem 2.12 is loose by more than an order of magnitude on real data, which is a finding about the bound worth reporting.

Remark S8 (Controls that can fail)*.* Controls 1–3 are chosen so that each can fail on well-formed input. Control 1 tests a theorem whose failure probability is bounded by α, so a failure is strong evidence against the computation or the error model. Control 2 tests an invariance that the theory asserts, so a failure localises to the implementation. Control 3 tests a bound rather than an identity, so a failure is informative about the bound's tightness. A control that cannot fail measures nothing, and we have excluded from the protocol several checks with that property — notably any check comparing the implementation's verdict against the same implementation's tolerance, which would pass by construction.

Outcome on the erythrocyte network

We executed the protocol on the human erythrocyte network assembled from UniProtKB/Swiss-Prot (The UniProt Consortium, 2025) and Rhea (Bansal et al., 2022) (main text, Section 2.7) rather than on the Recon3D glycolysis–TCA subnetwork named under *Data* above, since erythrocytes have no mitochondria and no tricarboxylic acid cycle. Step 3 fails: the pyruvate kinase reaction lies on none of the network's seven cycles, because no erythrocyte enzyme returns pyruvate to phosphoenolpyruvate. By the criterion of step 3 the experiment is invalid by design, and steps 4–10 were not run. No erythrocyte cycle has a measured equilibrium constant on every edge (Section S17), so on present data there is no substitute target either.

S17: analysis of experimental data

The code is `validation/realdata/run_realdata.py`; every number below is read from `results_real.json`.

Cycles of a stoichiometric network

Let $S \in \mathbb{Z}^{n\times m}$ be the stoichiometric matrix of m reactions over n species, after balancing with water and protons and removing the proton row. A cycle is an integer $z \in \ker S$; its sum on edge data $d \in \mathbb{R}^m$ is $H_z = z^\top d$. If $d = S^\top \mu$ for any $\mu \in \mathbb{R}^n$, then $H_z = (Sz)^\top \mu = 0$; this is Lemma 2.8, and with μ replaced by species-level errors it is Lemma 2.17. The proof of Theorem 2.19 uses only the linearity of H_z in d and so holds unchanged, with variance $\sum_e z_e^2 \sigma_e^2$. The incidence matrix of the main text is the case in which every column of S has one $+1$ and one -1.

Proposition S9 (Numerical floor for edge data)**.** *Let the edge data d be held in the arithmetic of Assumption 2.11, and let z have k nonzero entries, each exactly representable. Then the value $\widehat{H}_z$ of $z^\top d$ computed by forming the k products and summing them in any order satisfies*

$$|\widehat{H}_z - H_z| \;\le\; \gamma_k \sum_e |z_e d_e|.$$

Proof. This is the standard bound for an inner product of length k (Higham, 2002, Section 3.1): each product incurs at most one relative rounding error of size u, none when $|z_e|$ is a power of two, and any summation order incurs at most $k-1$ more along the path of each term. Collecting the factors $(1+\delta_j)$ by Lemma 3.1 of Higham (2002) gives the bound. When the data are differences of potentials, forming each difference adds one rounding per term, which is the source of γ_{2L} in Theorem 2.12. □

We span $\ker S$ by the fundamental basis of the reduced row-echelon form of S, computed in exact rational arithmetic and scaled to coprime integers, so that the basis carries no rounding error of its own.

Data, transform and aggregation

TECRDB as distributed with component contribution (commit `67d3804`) has 4,544 rows; 3,912 report K', T and pH. Of these, 173 involve a compound without a structure in the eQuilibrator cache and were dropped; every remaining reaction balanced with water and protons. Restricting to grades A–C and 293–313 K leaves 3,276 measurements from 908 studies of 418 distinct reactions. For a measurement of reaction e at conditions $c = (\mathrm{pH}, I, T, \mathrm{pMg})$,

$$d_e \;=\; -RT \ln K'_e \;-\; \sum_i S_{ie}\, \tau_i(c),$$

where $\tau_i(c)$ is the Legendre transform of compound i computed by eQuilibrator (Beber et al., 2022) from its pKa and magnesium-binding constants; a missing ionic strength is taken as 0.25 M and a missing pMg as 14, as component contribution does (Noor et al., 2013). Any reference-state term in τ_i is a species-level quantity and cancels around every cycle (Lemma 2.17), so only the dependence of τ_i on each measurement's own conditions reaches a cycle sum. As a check of the transform, the values recovered at pH 7, $I = 0.25$ M and 298.15 K are $+7.7$ kJ/mol for triose-phosphate isomerase (DHAP to GAP), -2.4 for adenylate kinase (ATP + AMP to 2 ADP), -3.0 for enolase (2PG to PEP) and $+3.2$ for glucose-6-phosphate isomerase (G6P to F6P), in line with tabulated standard values (Alberty, 1998).

Measurements repeated within one study are averaged into one value; a reaction's datum is the mean over its n_e studies, with $\sigma_e = \sigma/\sqrt{n_e}$ and σ the pooled between-study standard deviation, $\big(\sum_e \sum_s (x_{es} - \bar{x}_e)^2 / \sum_e (n_e - 1)\big)^{1/2} = 2.68$ kJ/mol over the 165 reactions measured in at least two studies (1.49 kJ/mol for grade A alone).

Calibration on repeated measurements

For a reaction measured in $n \ge 2$ studies, the leave-one-out deviation $t_s = (x_s - \bar{x}_{-s})/(\sigma\sqrt{1 + 1/(n-1)})$ is the standardised sum of a two-edge cycle whose true sum is zero. Over all grades the 655

values exceed the Gaussian quantile 2.576 in 3.5% of cases (nominal 1%); their 95th and 99th percentiles are 2.15 and 3.81, and the largest is 5.26. For grade A alone the 85 values exceed 2.576 in 1.2% of cases and their 99th percentile is 2.58. Across the 67 basis cycles of the main analysis the standardised sums $Z = H_z/(\sum_e z_e^2\sigma_e^2)^{1/2}$ have standard deviation 1.75, and 10.4% exceed 2.576.

Table S1: Verdicts on the 67 basis cycles of the measured-reaction network (grades A–C) and the 13 of the grade-A network, by the quantile used in the data floor, and the number flagged by a fixed tolerance. No cycle is CONSISTENT under any rule.

Data	Rule	INCONSISTENT	UNDECIDABLE
A–C	cycle-local, Gaussian (2.58)	7	60
A–C	cycle-local, replicate-calibrated (3.81)	3	64
A–C	cycle-local, Chebyshev (10)	0	67
A–C	fixed $\varepsilon = 10^{-6}$ or 10^{-3} kJ/mol	67	—
A–C	fixed $\varepsilon = 1$, 5, 10 kJ/mol	54, 27, 14	—
A	cycle-local, Gaussian or calibrated (2.58)	2	11
A	cycle-local, Chebyshev (10)	0	13

The flagged cycles

The seven cycles flagged under the Gaussian floor in the reference basis are listed by EC number, with each reaction's coefficient and the number of studies behind it; "&" marks a single measurement of a coupled reaction.

Table S2: Cycles flagged INCONSISTENT (Gaussian data floor, $\alpha = 0.01$) in the reference basis. H and ε^* in kJ/mol.

H	ε^*	Z	Reactions: coefficient, EC (studies)
−27.77	8.45	−8.46	+1 2.7.1.90 (1); −1 3.1.3.11 (3); +1 3.6.1.1 (6)
−13.77	8.45	−4.20	+1 2.7.7.4 (3); −1 2.7.7.4&3.6.1.1 (1); +1 3.6.1.1 (6)
23.97	16.18	3.82	+1 2.1.3.1 (1); −1 3.6.1.38 (1); −1 1.1.1.37&4.1.3.7 (1); +1 1.1.1.39 (1); +1 4.1.3.7 (1); +1 6.4.1.3 (2)
14.51	11.34	3.30	+1 1.5.1.15 (1); −1 1.4.1.2 (5); −1 1.5.1.5 (1); +1 1.4.1.3 (4); +1 3.5.4.9 (4)
−14.51	12.28	−3.04	+1 4.1.2.13 (2); +1 3.1.3.1 (2); −1 3.1.3.2 (1); −1 2.2.1.2 (1); −1 4.1.2.13 (6)
−10.02	9.63	−2.68	+1 1.1.1.3 (2); −1 1.4.1.2 (5); +1 1.4.1.3 (4); +1 1.1.1.3 (1)
20.03	19.52	2.64	+1 2.7.7 (1); +1 3.6.1.38 (1); +1 1.1.1.37&4.1.3.7 (1); +1 4.1.1.38 (1); −1 1.1.1.39 (1); −1 4.1.3.7 (1); +1 3.1.4 (1); −1 2.7.9.1 (3); −2 3.6.1.1 (6)

A repeated EC number within a cycle marks distinct reactions of one enzyme class (for example, glutamate dehydrogenase with NAD^+ and with $NADP^+$). Over 200 further fundamental bases, from random column orders, some cycle was flagged in every basis, the number flagged ranged from 2 to 40, and the intersection of flagged cycles was always empty. The reactions lying on a flagged cycle in at least 75% of bases are pyrophosphate–fructose-6-phosphate 1-phosphotransferase (EC 2.7.1.90, 94%), methylmalonyl-CoA carboxyltransferase (2.1.3.1, 88.5%), propanoyl-CoA carboxylase (6.4.1.3, 88.5%), methylenetetrahydrofolate dehydrogenase (NAD^+; 1.5.1.15, 80.5%) and ADP-glucose phosphodiesterase (3.1.4, 78%); all but one rest on a single study.

The erythrocyte network

The 38 genes are HK1, GPI, PFKL, PFKM, ALDOA, TPI1, GAPDH, PGK1, BPGM, MINPP1, PGAM1, ENO1, PKLR, LDHA, LDHB, G6PD, PGLS, PGD, RPE, RPIA, TKT, TALDO1, PGM1, AK1, NME1, NME2, GSR, GPX1, GCLC, GSS, ADA, AMPD3, PNP, APRT, HPRT1, PRPS1, ATP1A1 and ATP2B4. Their UniProtKB/Swiss-Prot entries annotate 118 Rhea reactions. We excluded 27 whose participants are compound classes without a defined structure (such as "a D-hexose"), 8 with generic macromolecular participants, and 9 with no net chemistry once compartments are merged (pure transport), leaving 74, of which 70 are chemically distinct, over 101 species. S has rank 63, so $\nu = 7$, and 19 reactions lie on a cycle.

One basis cycle is the Rapoport–Luebering shunt: phosphoglycerate kinase (3 studies), phosphoglycerate mutase (12), bisphosphoglycerate mutase (0), bisphosphoglycerate phosphatase (0) and ATP hydrolysis by the Ca^{2+} pump (1). The other six are cycles of nucleotide interconversion by adenylate kinase and nucleoside-diphosphate kinase on (deoxy)nucleotides, with at most one measured edge each. No cycle has a measured constant on every edge, and the 31 erythrocyte reactions with any measurement form a subnetwork with no cycle at all. Matching to TECRDB is at the level of molecular connectivity, so an anomer-specific Rhea reaction is matched to TECRDB's generic one.

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